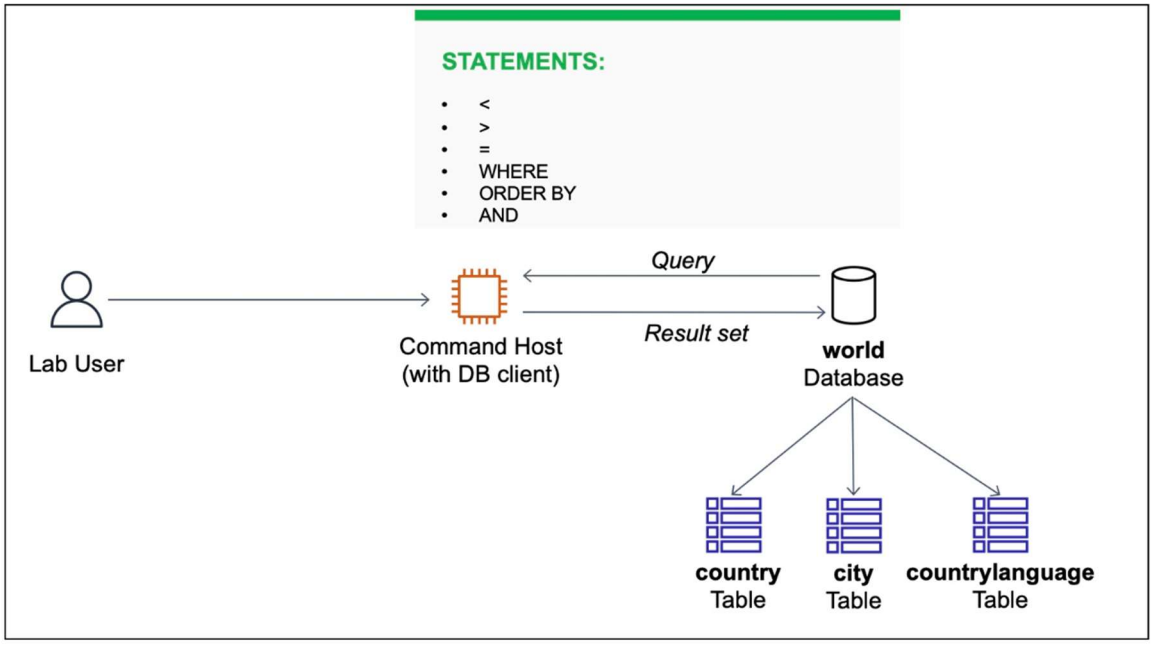
270--Lab - Selección de datos de una base de datos

En esta práctica se muestra el funcionamiento de los siguientes comandos en SQL:

- SELECT para hacer consultas a las tablas
- COUNT() para contar los registros de una tabla
- WHERE, AND y operadores de comparación para filtrar el resultado de la consulta
- ORDER BY para ordenar el resultado de la consulta

Escenario

Se cuenta con una instancia EC2 configurada con un cliente de base de datos MySQL.



Para conectarnos a la instancia y a la base de datos seguimos los pasos de las prácticas anteriores.

Hacer consultas a la base de datos

Verificamos que la base de datos *world* está disponible

```
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
| world              |
+--------------------+
4 rows in set (0.001 sec)
```

Verificamos el número de registros de la tabla *country*

```
MariaDB [(none)]> SELECT COUNT(*) FROM world.country;
+----------+
| COUNT(*) |
+----------+
|      239 |
+----------+
1 row in set (0.003 sec)
```

Checamos las columnas de la tabla *country*

```
MariaDB [(none)]> SHOW COLUMNS FROM world.country;
+---------------+-----------------------------------------------------------------------------------------+------+-----+---------+-------+
| Field         | Type                                                                                    | Null | Key | Default | Extra |
+---------------+-----------------------------------------------------------------------------------------+------+-----+---------+-------+
| Code          | char(3)                                                                                 | NO   | PRI |         |       |
| Name          | char(52)                                                                                | NO   |     |         |       |
| Continent     | enum('Asia','Europe','North America','Africa','Oceania','Antarctica','South America') | NO   |     | Asia    |       |
| Region        | char(26)                                                                                | NO   |     |         |       |
| SurfaceArea   | decimal(10,2)                                                                           | NO   |     | 0.00    |       |
| IndepYear     | smallint(6)                                                                             | YES  |     | NULL    |       |
| Population    | int(11)                                                                                 | NO   |     | 0       |       |
| LifeExpectancy | decimal(3,1)                                                                            | YES  |     | NULL    |       |
| GNP           | decimal(10,2)                                                                           | YES  |     | NULL    |       |
| GNPOld        | decimal(10,2)                                                                           | YES  |     | NULL    |       |
| LocalName     | char(45)                                                                                | NO   |     |         |       |
| GovernmentForm | char(45)                                                                                | NO   |     |         |       |
| Capital       | int(11)                                                                                 | YES  |     | NULL    |       |
| Code2         | char(2)                                                                                 | NO   |     |         |       |
+---------------+-----------------------------------------------------------------------------------------+------+-----+---------+-------+
14 rows in set (0.001 sec)
```

La diferencia entre DESCRIBE y SHOW COLUMNS FROM es que el segundo es más flexible porque permite filtrar los resultados usando una condición con la cláusula WHERE. Por ejemplo, podemos consultar cuales son las columnas de tipo entero:

```
MariaDB [(none)]> SHOW COLUMNS FROM world.country  WHERE Type LIKE '%int%';
+------------+-------------+------+-----+---------+-------+
| Field      | Type        | Null | Key | Default | Extra |
+------------+-------------+------+-----+---------+-------+
| IndepYear  | smallint(6) | YES  |     | NULL    |       |
| Population | int(11)     | NO   |     | 0       |       |
| Capital    | int(11)     | YES  |     | NULL    |       |
+------------+-------------+------+-----+---------+-------+
3 rows in set (0.001 sec)
```

Hacemos una consulta a la tabla *country* mostrando solo ciertas columnas

```
MariaDB [(none)]> SELECT Name, Capital, Region, SurfaceArea, Population FROM world.country;
+---------------------------------------------+---------+---------------------------+-------------+------------+
| Name                                        | Capital | Region                    | SurfaceArea | Population |
+---------------------------------------------+---------+---------------------------+-------------+------------+
| Aruba                                       |     129 | Caribbean                 |      193.00 |     103000 |
| Afghanistan                                 |       1 | Southern and Central Asia |   652090.00 |   22720000 |
| Angola                                      |      56 | Central Africa            |  1246700.00 |   12878000 |
| Anguilla                                    |      62 | Caribbean                 |       96.00 |       8000 |
| Albania                                     |      34 | Southern Europe           |    28748.00 |    3401200 |
| Andorra                                     |      55 | Southern Europe           |      468.00 |      78000 |
| Netherlands Antilles                        |      33 | Caribbean                 |      800.00 |     217000 |
| United Arab Emirates                        |      65 | Middle East               |    83600.00 |    2441000 |
| Argentina                                   |      69 | South America             |  2780400.00 |   37032000 |
| Armenia                                     |     126 | Middle East               |    29800.00 |    3520000 |
| American Samoa                              |      54 | Polynesia                 |      199.00 |      68000 |
| Antarctica                                  |    NULL | Antarctica                | 13120000.00 |          0 |
| French Southern territories                 |    NULL | Antarctica                |     7780.00 |          0 |
```

Con *AS* indicamos un alias ("Surface Area") para la columna *SurfaceArea*

```
MariaDB [(none)]> SELECT Name, Capital, Region, SurfaceArea AS "Surface Area", Population FROM world.country;
+---------------------------------------------+---------+---------------------------+--------------+------------+
| Name                                        | Capital | Region                    | Surface Area | Population |
+---------------------------------------------+---------+---------------------------+--------------+------------+
| Aruba                                       |     129 | Caribbean                 |       193.00 |     103000 |
| Afghanistan                                 |       1 | Southern and Central Asia |    652090.00 |   22720000 |
| Angola                                      |      56 | Central Africa            |   1246700.00 |   12878000 |
| Anguilla                                    |      62 | Caribbean                 |         96.00 |       8000 |
| Albania                                     |      34 | Southern Europe           |      28748.00 |    3401200 |
| Andorra                                     |      55 | Southern Europe           |        468.00 |      78000 |
| Netherlands Antilles                        |      33 | Caribbean                 |        800.00 |     217000 |
| United Arab Emirates                        |      65 | Middle East               |      83600.00 |    2441000 |
| Argentina                                   |      69 | South America             |    2780400.00 |   37032000 |
| Armenia                                     |     126 | Middle East               |      29800.00 |    3520000 |
| American Samoa                              |      54 | Polynesia                 |        199.00 |      68000 |
| Antarctica                                  |    NULL | Antarctica                |   13120000.00 |          0 |
| French Southern territories                 |    NULL | Antarctica                |       7780.00 |          0 |
```

Con ORDER BY ordenamos el resultado de la consulta por la columna *Population* en orden ascendente

```
MariaDB [(none)]> SELECT Name, Capital, Region, SurfaceArea AS "Surface Area", Population FROM world.country ORDER BY Population;
+---------------------------------------------+---------+---------------------------+--------------+------------+
| Name                                        | Capital | Region                    | Surface Area | Population |
+---------------------------------------------+---------+---------------------------+--------------+------------+
| French Southern territories                 |    NULL | Antarctica                |      7780.00 |          0 |
| Heard Island and McDonald Islands           |    NULL | Antarctica                |       359.00 |          0 |
| United States Minor Outlying Islands        |    NULL | Micronesia/Caribbean      |        16.00 |          0 |
| South Georgia and the South Sandwich Islands |    NULL | Antarctica                |      3903.00 |          0 |
| Bouvet Island                               |    NULL | Antarctica                |        59.00 |          0 |
| British Indian Ocean Territory              |    NULL | Eastern Africa            |        78.00 |          0 |
| Antarctica                                  |    NULL | Antarctica                |  13120000.00 |          0 |
| Pitcairn                                    |    2912 | Polynesia                 |        49.00 |         50 |
| Cocos (Keeling) Islands                     |    2317 | Australia and New Zealand |        14.00 |        600 |
| Holy See (Vatican City State)               |    3538 | Southern Europe           |         0.40 |       1000 |
```

Ahora ordenamos el resultado por la columna *Population* en orden descendente

```
MariaDB [(none)]> SELECT Name, Capital, Region, SurfaceArea AS "Surface Area", Population FROM world.country ORDER BY Population DESC;
+-----------------------------------------+---------+-------------------------+--------------+------------+
| Name                                    | Capital | Region                  | Surface Area | Population |
+-----------------------------------------+---------+-------------------------+--------------+------------+
| China                                   |    1891 | Eastern Asia            |   9572900.00 | 1277558000 |
| India                                   |    1109 | Southern and Central Asia |  3287263.00 | 1013662000 |
| United States                           |    3813 | North America           |   9363520.00 |  278357000 |
| Indonesia                               |     939 | Southeast Asia          |   1904569.00 |  212107000 |
| Brazil                                  |     211 | South America           |   8547403.00 |  170115000 |
| Pakistan                                |    2831 | Southern and Central Asia |   796095.00 |  156483000 |
| Russian Federation                      |    3580 | Eastern Europe          |  17075400.00 |  146934000 |
| Bangladesh                              |     150 | Southern and Central Asia |  143998.00 |  129155000 |
| Japan                                   |    1532 | Eastern Asia            |    377829.00 |  126714000 |
| Nigeria                                 |    2754 | Western Africa          |    923768.00 |  111506000 |
| Mexico                                  |    2515 | Central America         |   1958201.00 |   98881000 |
```

Agregamos un *WHERE* a la consulta para filtrar por la columna *Population* mayor a 50,000,000

```
MariaDB [(none)]> SELECT Name, Capital, Region, SurfaceArea AS "Surface Area", Population FROM world.country WHERE Population > 50000000 ORDER BY Population DESC;
+-------------------------------------+---------+-------------------------+--------------+------------+
| Name                                | Capital | Region                  | Surface Area | Population |
+-------------------------------------+---------+-------------------------+--------------+------------+
| China                               |    1891 | Eastern Asia            |   9572900.00 | 1277558000 |
| India                               |    1109 | Southern and Central Asia |  3287263.00 | 1013662000 |
| United States                       |    3813 | North America           |   9363520.00 |  278357000 |
| Indonesia                           |     939 | Southeast Asia          |   1904569.00 |  212107000 |
| Brazil                              |     211 | South America           |   8547403.00 |  170115000 |
| Pakistan                            |    2831 | Southern and Central Asia |   796095.00 |  156483000 |
| Russian Federation                  |    3580 | Eastern Europe          |  17075400.00 |  146934000 |
| Bangladesh                          |     150 | Southern and Central Asia |  143998.00 |  129155000 |
| Japan                               |    1532 | Eastern Asia            |    377829.00 |  126714000 |
| Nigeria                             |    2754 | Western Africa          |    923768.00 |  111506000 |
| Mexico                              |    2515 | Central America         |   1958201.00 |   98881000 |
| Germany                             |    3068 | Western Europe          |    357022.00 |   82164700 |
| Vietnam                             |    3770 | Southeast Asia          |    331689.00 |   79832000 |
| Philippines                         |     766 | Southeast Asia          |    300000.00 |   75967000 |
| Egypt                               |     608 | Northern Africa         |   1001449.00 |   68470000 |
```

Ahora filtramos el resultado por la columna *Population* mayor a 50,000,000 y menor a 100,000,000

```
MariaDB [(none)]> SELECT Name, Capital, Region, SurfaceArea AS "Surface Area", Population FROM world.country WHERE Population > 50000000 AND Population < 100000000 ORDER BY
Population DESC;
+-------------------------------------+---------+-------------------------+--------------+------------+
| Name                                | Capital | Region                  | Surface Area | Population |
+-------------------------------------+---------+-------------------------+--------------+------------+
| Mexico                              |    2515 | Central America         |   1958201.00 |   98881000 |
| Germany                             |    3068 | Western Europe          |    357022.00 |   82164700 |
| Vietnam                             |    3770 | Southeast Asia          |    331689.00 |   79832000 |
| Philippines                         |     766 | Southeast Asia          |    300000.00 |   75967000 |
| Egypt                               |     608 | Northern Africa         |   1001449.00 |   68470000 |
| Iran                                |    1380 | Southern and Central Asia |  1648195.00 |   67702000 |
| Turkey                              |    3358 | Middle East             |    774815.00 |   66591000 |
| Ethiopia                            |     756 | Eastern Africa          |   1104300.00 |   62565000 |
| Thailand                            |    3320 | Southeast Asia          |    513115.00 |   61399000 |
| United Kingdom                      |     456 | British Islands         |    242900.00 |   59623400 |
| France                              |    2974 | Western Europe          |    551500.00 |   59225700 |
| Italy                               |    1464 | Southern Europe         |    301316.00 |   57680000 |
| Congo, The Democratic Republic of the |  2298 | Central Africa          |   2344858.00 |   51654000 |
| Ukraine                             |    3426 | Eastern Europe          |    603700.00 |   50456000 |
+-------------------------------------+---------+-------------------------+--------------+------------+
14 rows in set (0.000 sec)
```

Por último, filtramos la consulta por la columna *Population* mayor a 50,000,000 y la región igual a "Southern Europe"

```
MariaDB [(none)]> SELECT Name, Capital, Region, SurfaceArea AS "Surface Area", Population FROM world.country WHERE Population > 50000000 AND Region='Southern Europe';
+-------+---------+----------------+--------------+------------+
| Name  | Capital | Region         | Surface Area | Population |
+-------+---------+----------------+--------------+------------+
| Italy |    1464 | Southern Europe |    301316.00 |   57680000 |
+-------+---------+----------------+--------------+------------+
1 row in set (0.000 sec)
```